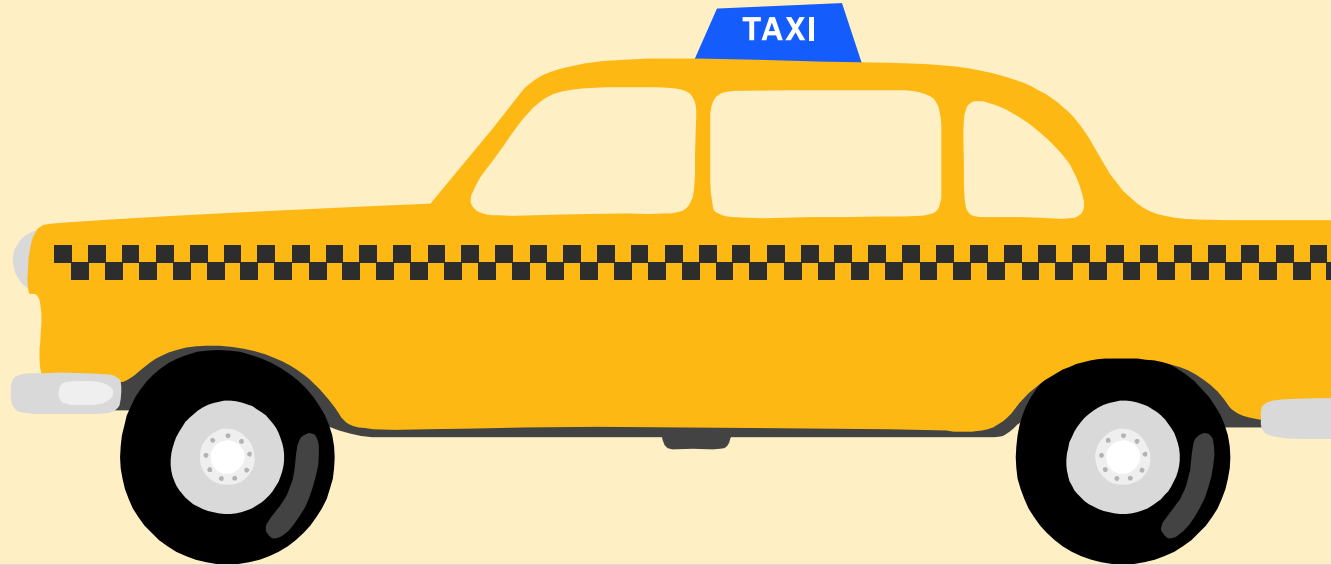
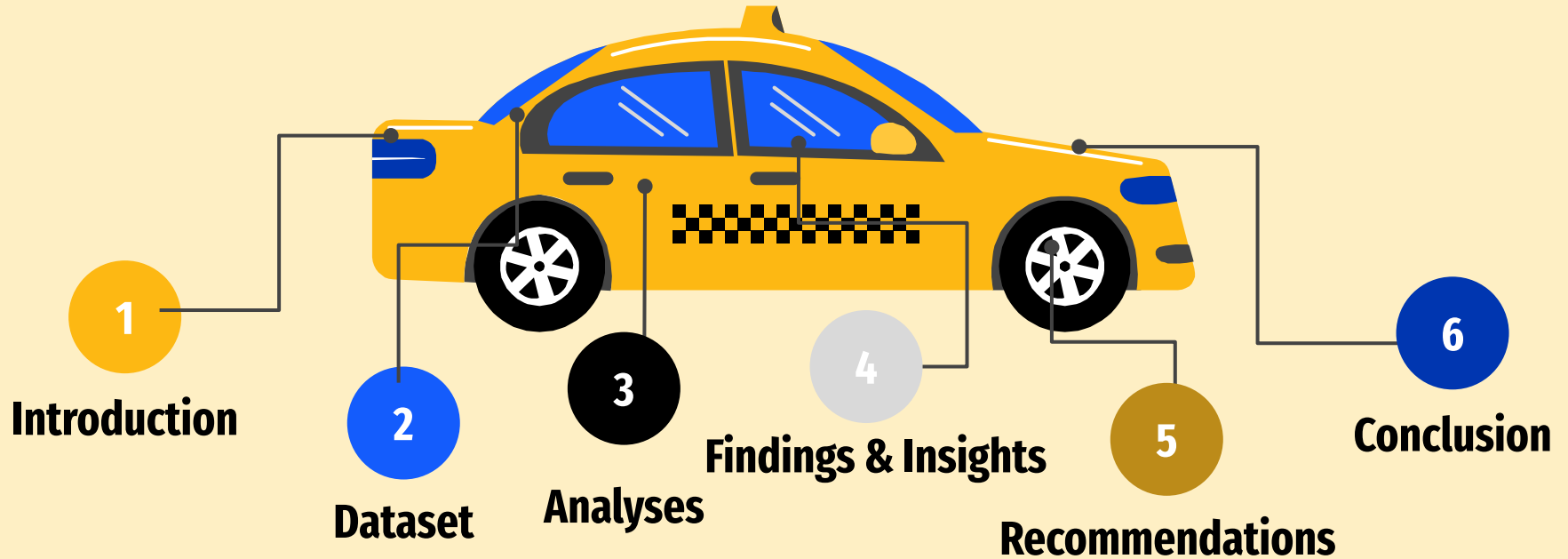


NYC Taxi Cab

By Brianna, Vanessa and Manavi



Agenda



What's Our Dataset?

Dataset Overview

We used a NYC yellow taxi dataset from Kaggle created by *Elemento*

It includes over 1 million rows and 19 columns

This dataset is created for the years of 2015-2016, we decided to focus in on 2016

The two vendors in this dataset are Creative Mobile Technologies (CMT) and Curb Mobility (Curb)



Important Variables

Trip Distance

Both vendors total trip distance over an amount of time in 2016



Total Amount

Both vendors total amount charged including all variables



Passenger Count

Both vendors overall passenger count



Payment Types

How customers paid for their Taxi service



Data Cleaning

Step 1



Primary Key

Added primary key for "Trip ID" to be able to identify the different trips

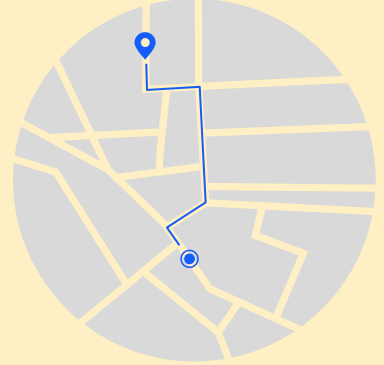
Step 2



Missing Values

Checked all columns for blank cells and found none.

Step 3



Summary Statistics

Used mean, median, and mode to catch outliers or unrealistic values in distance, fare, passenger count, etc

Data Cleaning

Step 4



Decoded

Translated numeric codes into descriptive labels for clarity

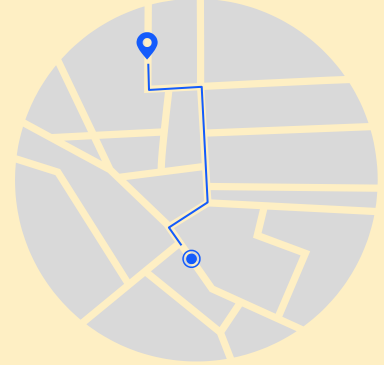
Step 5



Formatting

Reformatted (numbers, dates, and coordinates)

Step 6

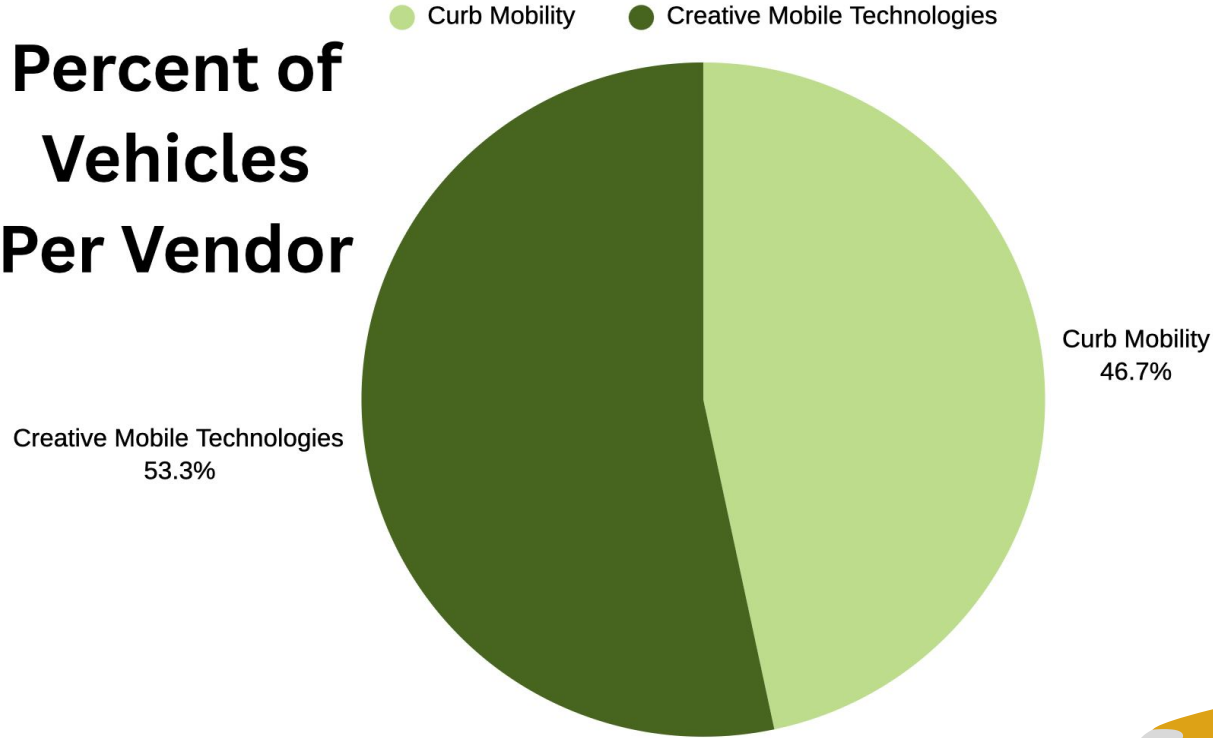


Visual Validation

Used bar charts and trend lines to validate data patterns, making sure values made sense

Vendor Data

Percent of Vehicles Per Vendor





Basic Infographics



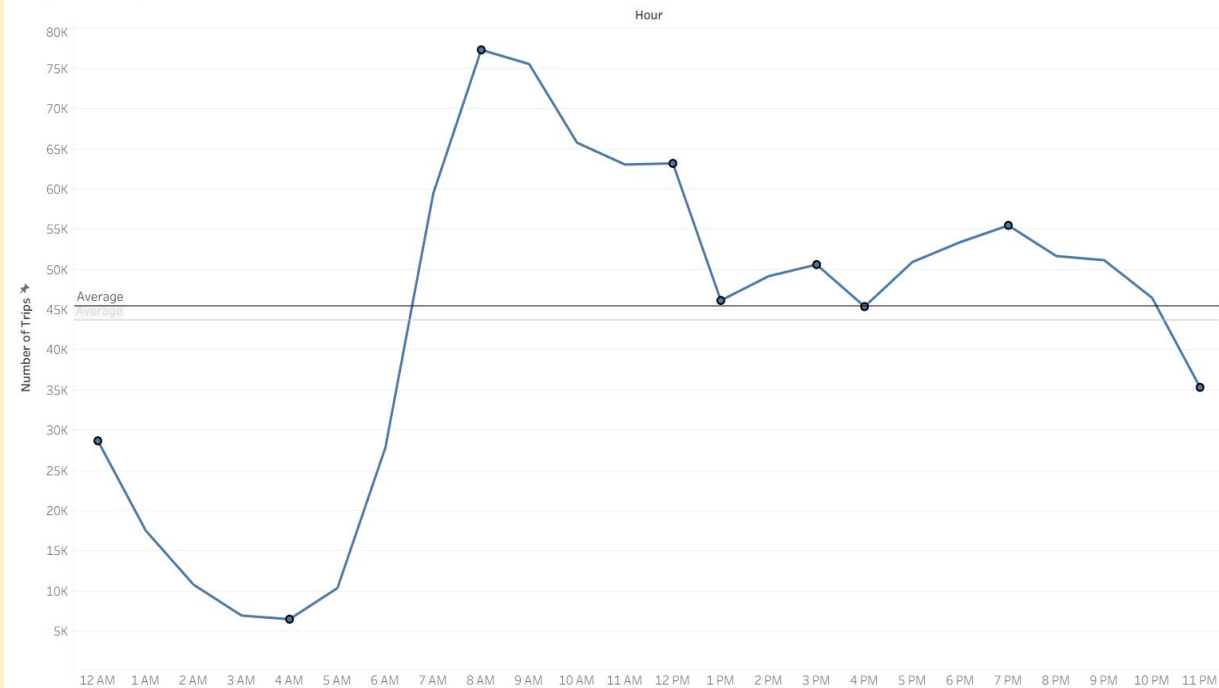
Columns

Hour

Rows

CNT(Hour in the Day)

DropOffTimeByHour



0 h

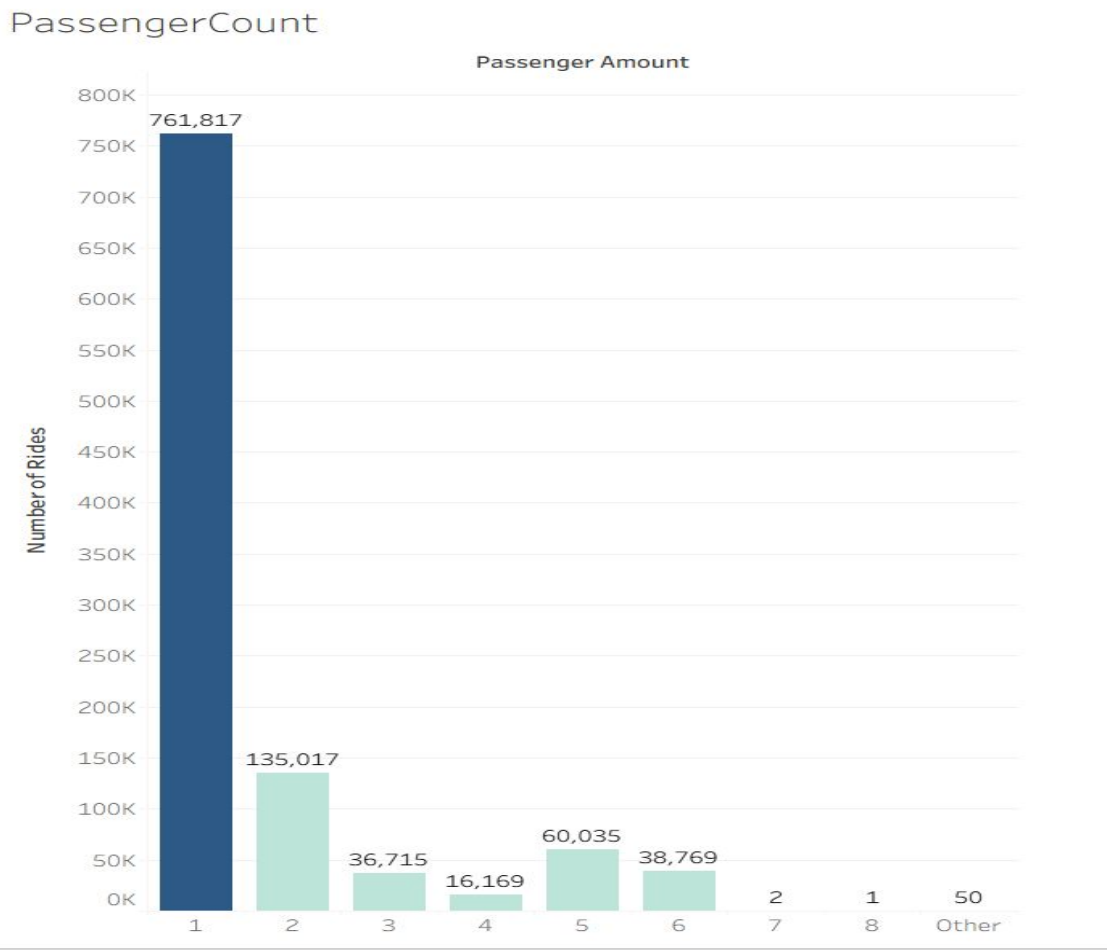
4am

8am

24 h
8

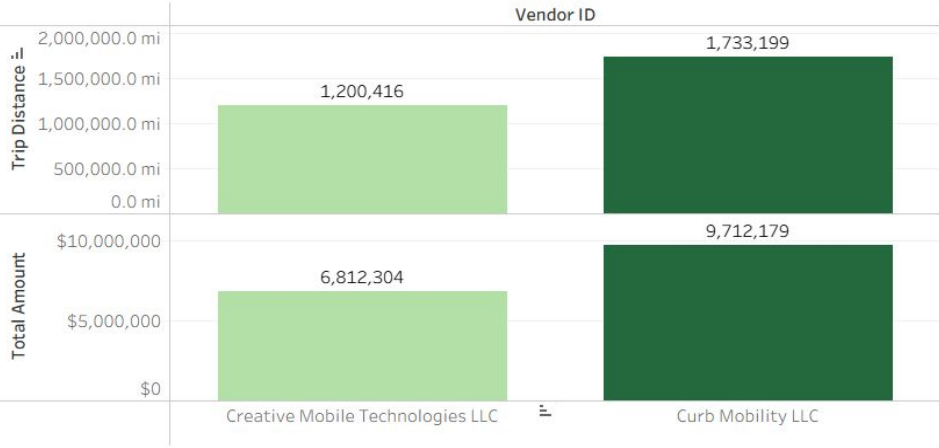


Basic Infographics

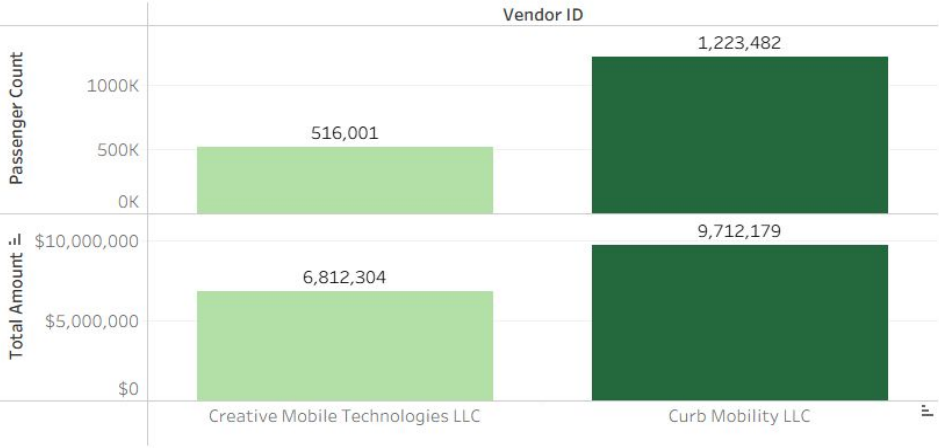


Creative Mobile Technologies LLC vs Curb Mobility LLC

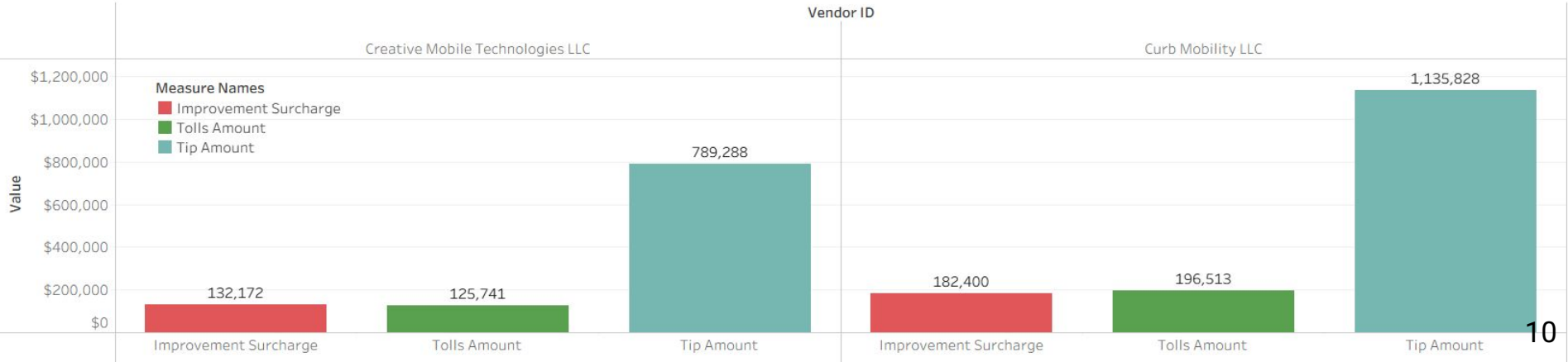
Total amount by trip distance by vendor



Total amount by passenger count by vendor



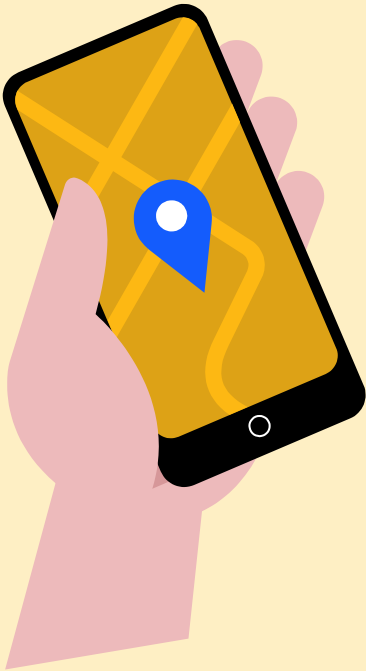
Breakdown of Total Amount



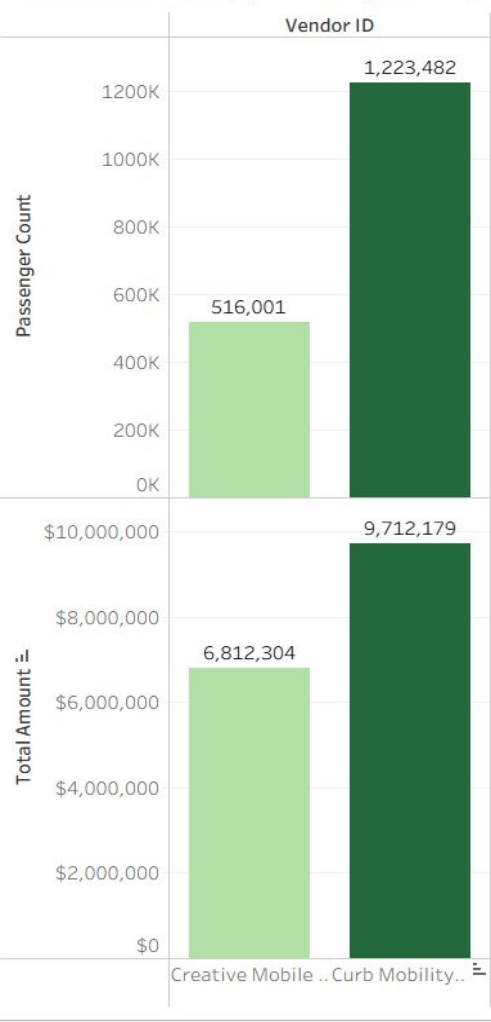
Breakdown of Total Amount



Total amount by trip distance by vendor



Total amount by passenger count by vendor



Recommendations



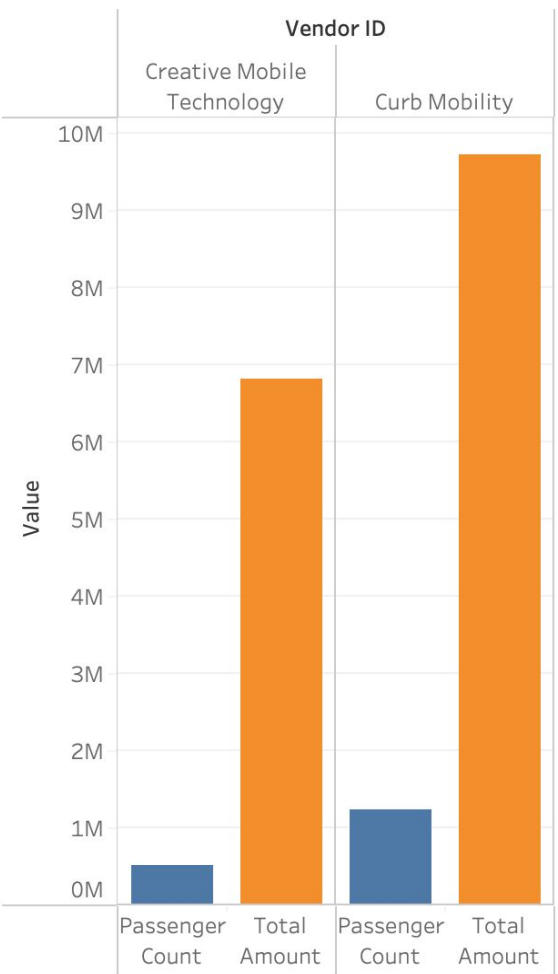
Creative

- ★ A recommendation for creative is to target longer routes, this will generate more revenue
- ★ Identify vehicles with few or short trips and reposition or reassign them based on data
- ★ Collaborate with airports, hotels, tourist spots to secure longer trip demand
- ★ Upgrade routing and matching systems so drivers get faster, higher value trip assignments and to compete with Curb's system

People starting a Taxi Business

- ★ Choose high demand areas
- ★ Focus on peak hours (rush hour, weekends, evenings)
- ★ Pick a strong technology vendor (using a reliable system will boost trip volume and minimize idle time)

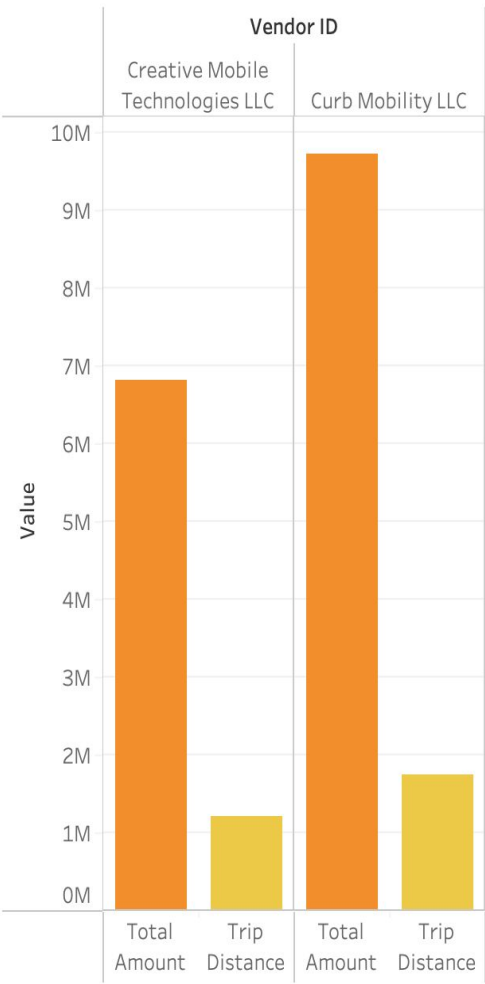
TotalPassengerCountByVendor



Basic Infographics



TotalAmountByVendor



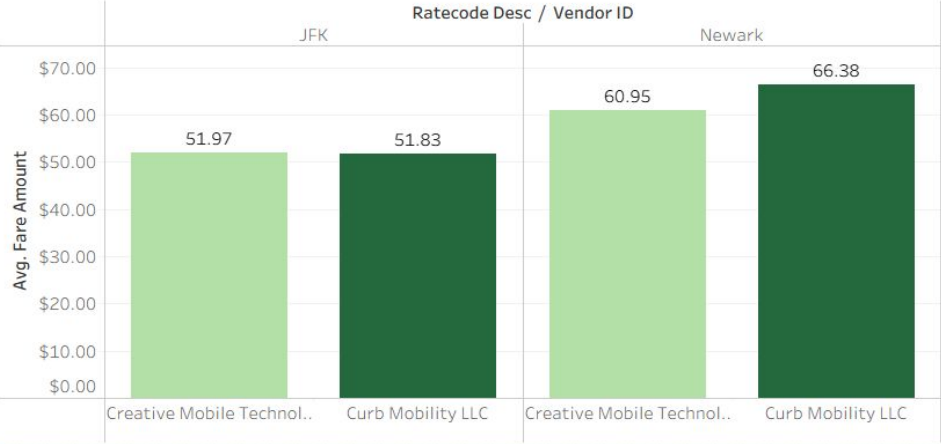
Vendor ID

Creative Mobile Technologies LLC

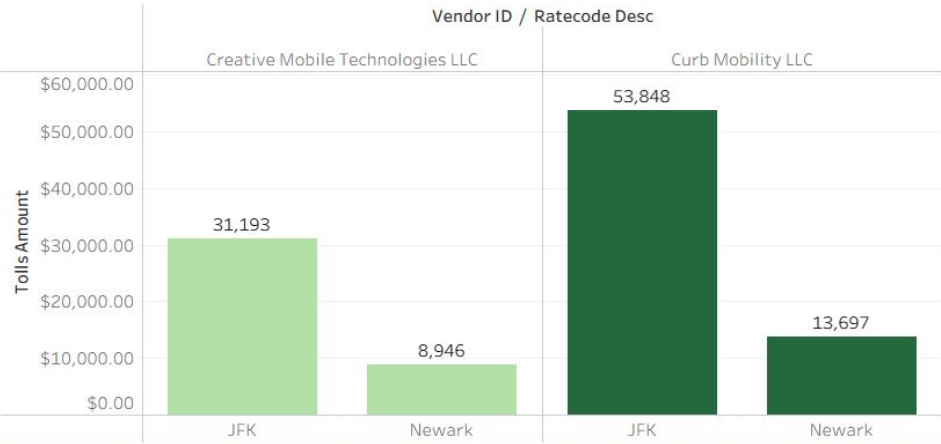
Curb Mobility LLC

Airport Insights

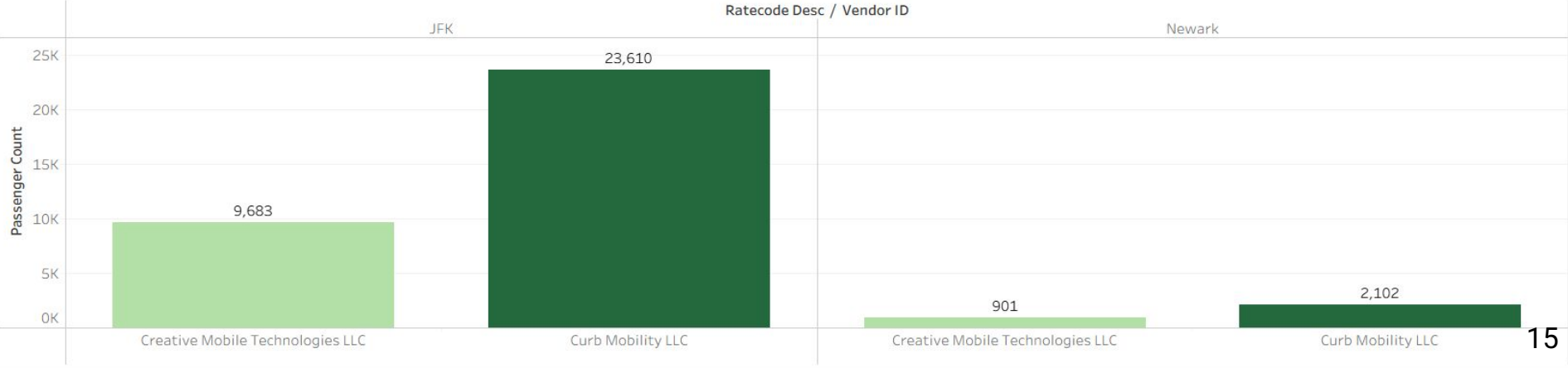
Avg Fare by Vendor & Airport



Tolls by Vendor & Airport



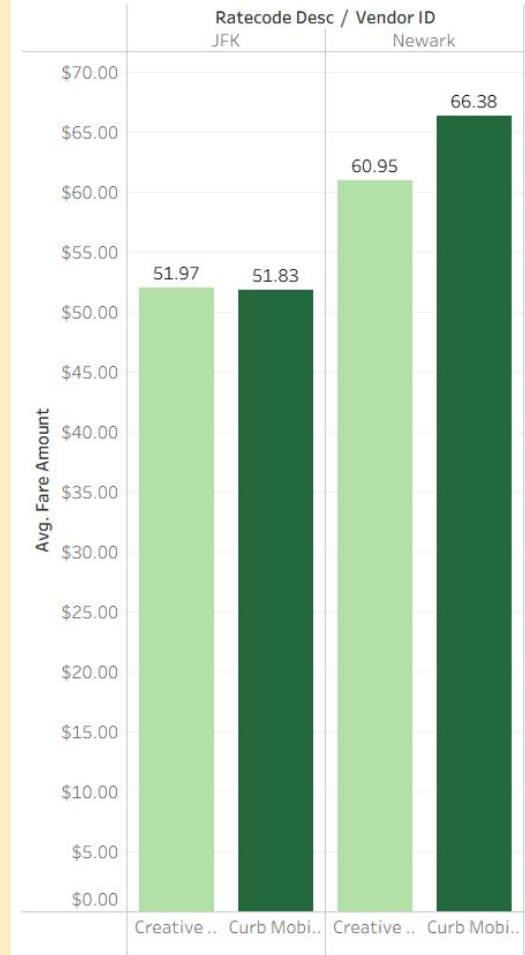
Passenger Amt by Vendor & Airport



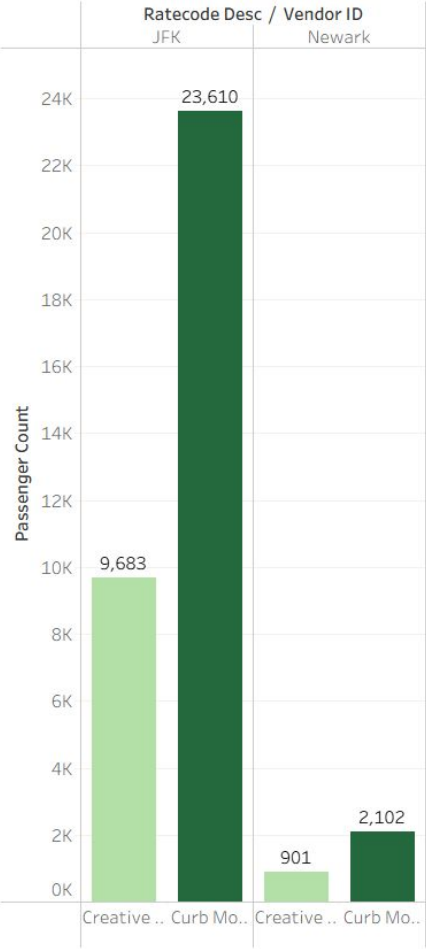
Tolls by Vendor & Airport



Avg Fare by Vendor & Airport



Passenger Amt by Vendor & Airport



Recommendations

To both Vendors:

- ★ Completely eliminate or focus less on going to Newark airport for passenger pickups
- ★ We also found that JFK has far more flights which causes people to choose this airport
- ★ Both vendors could slightly decrease their overall fare amount to increase overall pickups

Creative Mobile Technologies:

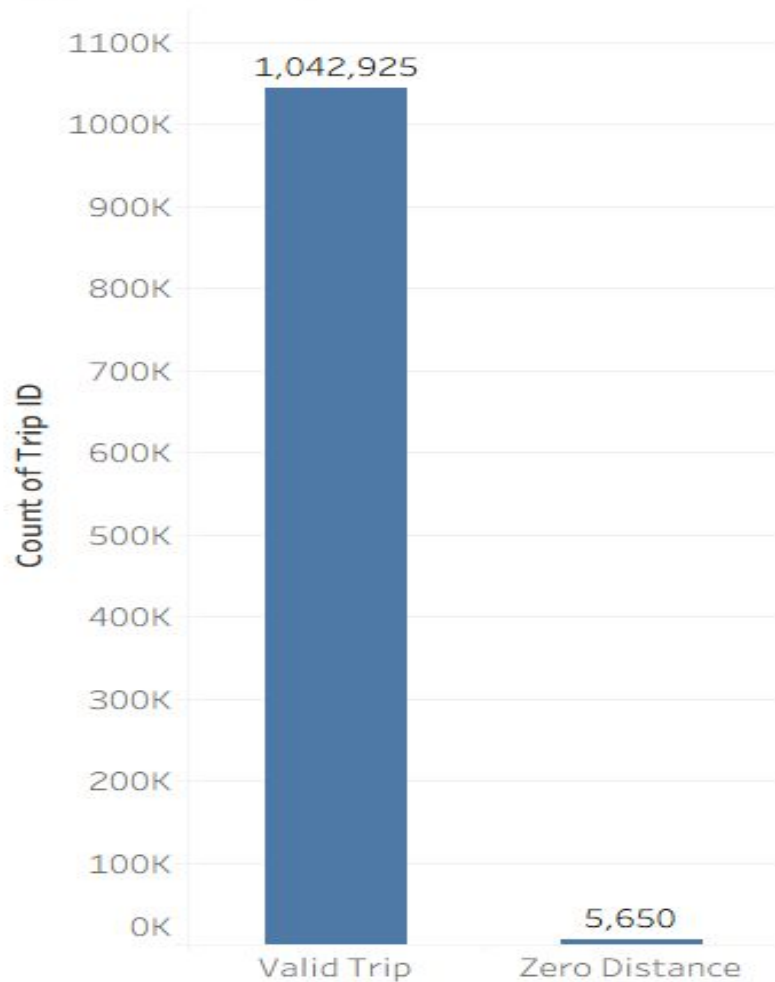
- ★ Should focus more on advertisements in order to increase their overall passenger count to JFK because Curb Mobility has roughly double the amount.

Passenger Tip:

- ★ We found that both vendors charge the same fare amount for JFK so both are a good option to pick.



Types of trips



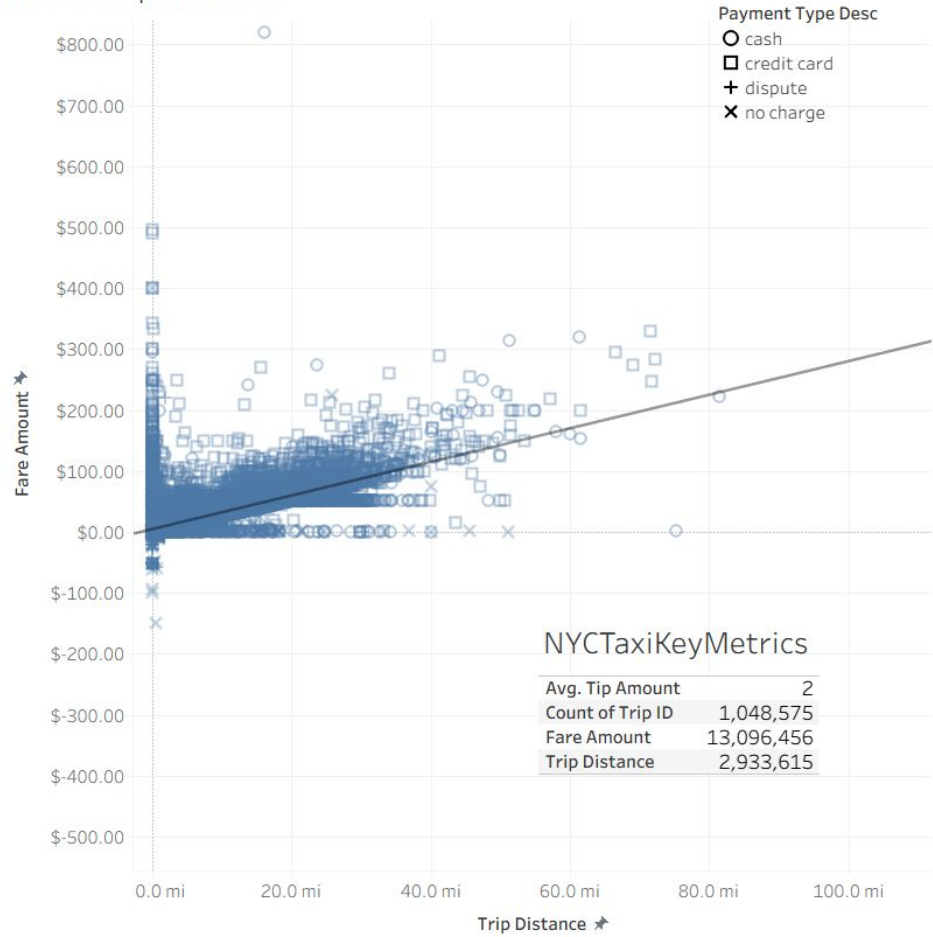
BASIC INFOGRAPHIC

- Valid trips - 1,042,925 trips
- Zero distance - 5,650 trips
- Zero distance/Valid trips ratio - 0.54

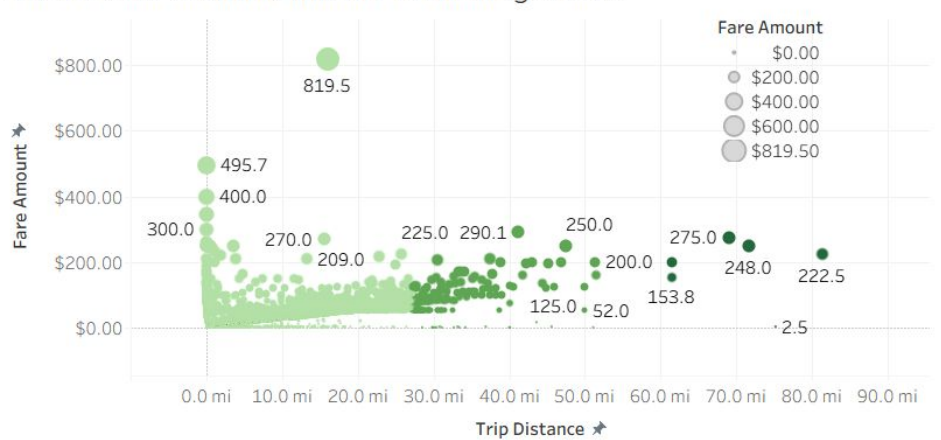
The ratio is small but does it not matter at all ?

Skeletons in the Closet

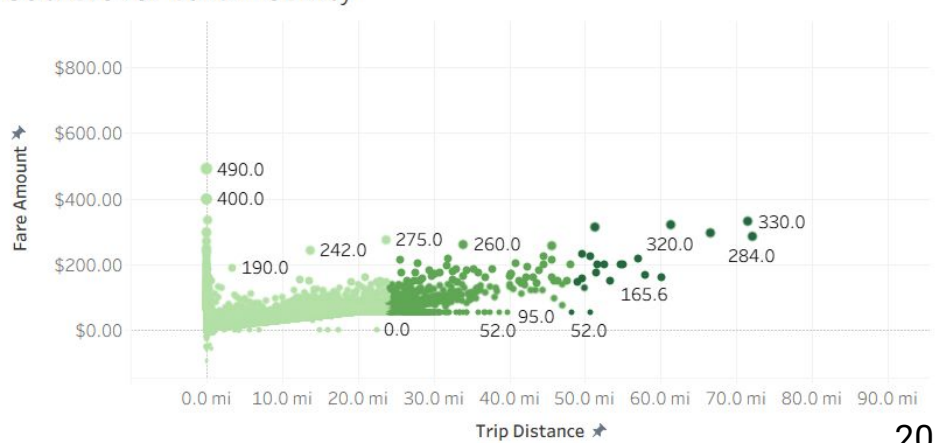
FareVSTripDistance



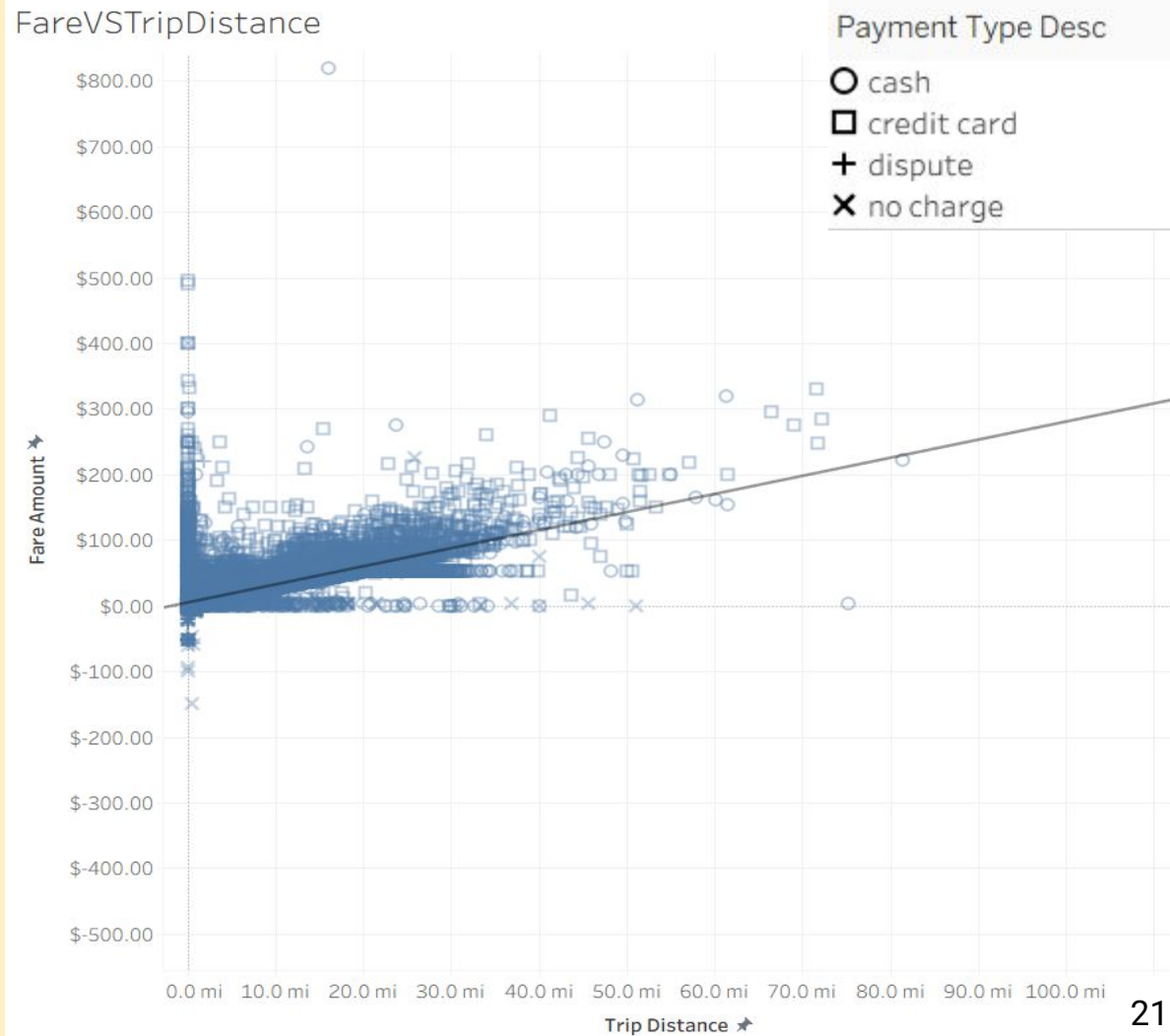
Outliers for Creative Mobile Technologies LLC



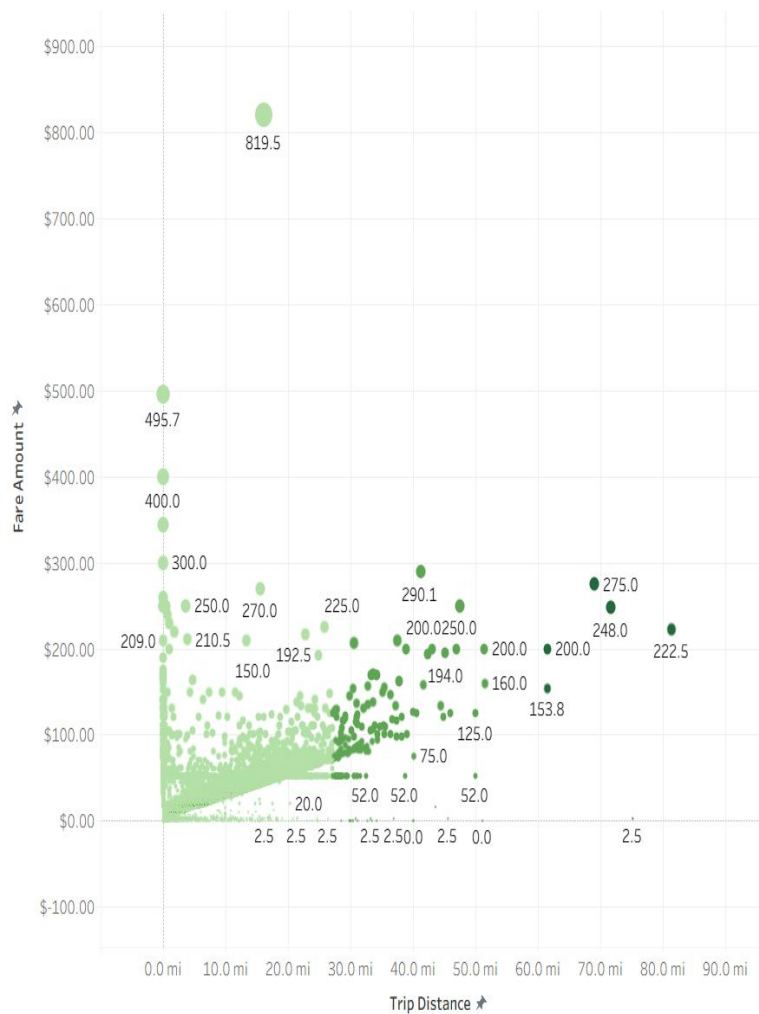
Outliers for Curb Mobility



Relationship between Fare Amount and Trip Distance with Payment Type description

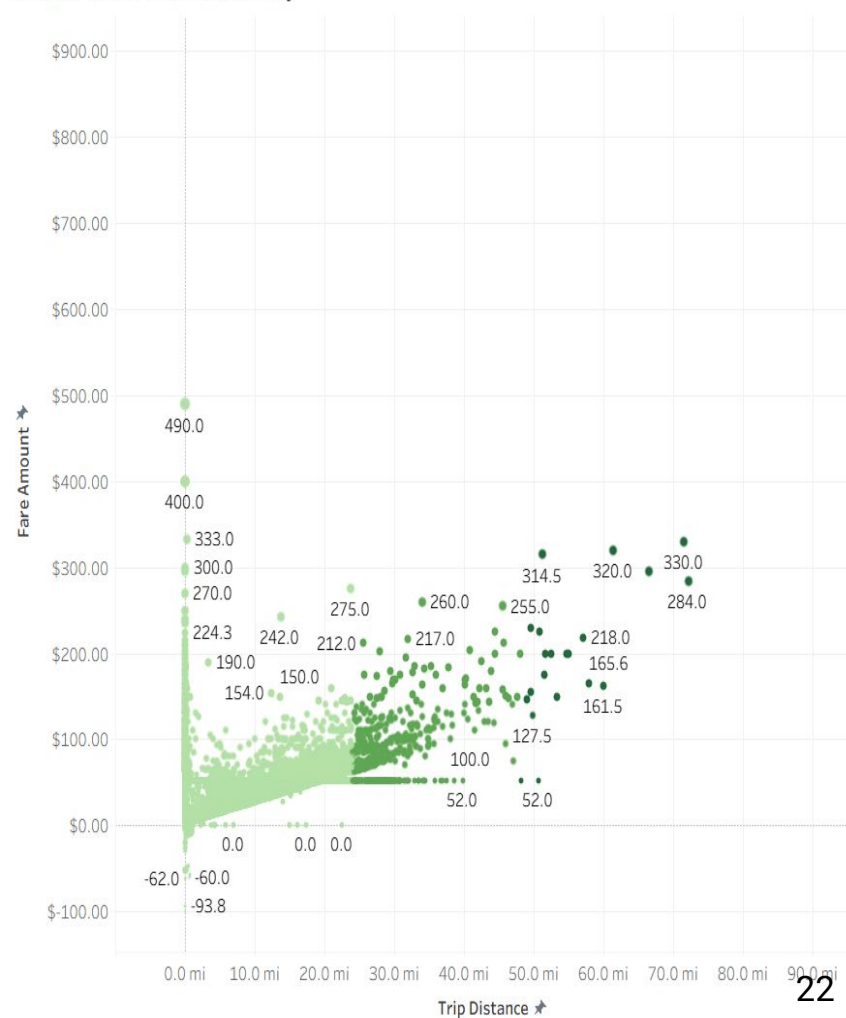


Outliers for Creative Mobile Technologies LLC



Creative
Mobile
tech
VS
Curb
Mobility.

Outliers for Curb Mobility



Recommendations

1. Upgrade GPS Tracking system and metering systems
2. Consider zero distance outliers as technical errors.
3. Always adapt to the latest systems and technical methods like Partnership between Creative Mobile Technologies and Uber

Conclusion

- Highlights comparative insights between Curb Mobility and Creative Mobile Technologies.
- Upgrading GPS systems and updated digital payment system can enhance routing and reduce transactional errors.
- Airport focused pickup and dropoff schedule for higher fare opportunities.
- Enhanced customer experience while promoting long term sustainability

References

How Much To Tip Taxi Drivers In 2025: And Why You Should

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